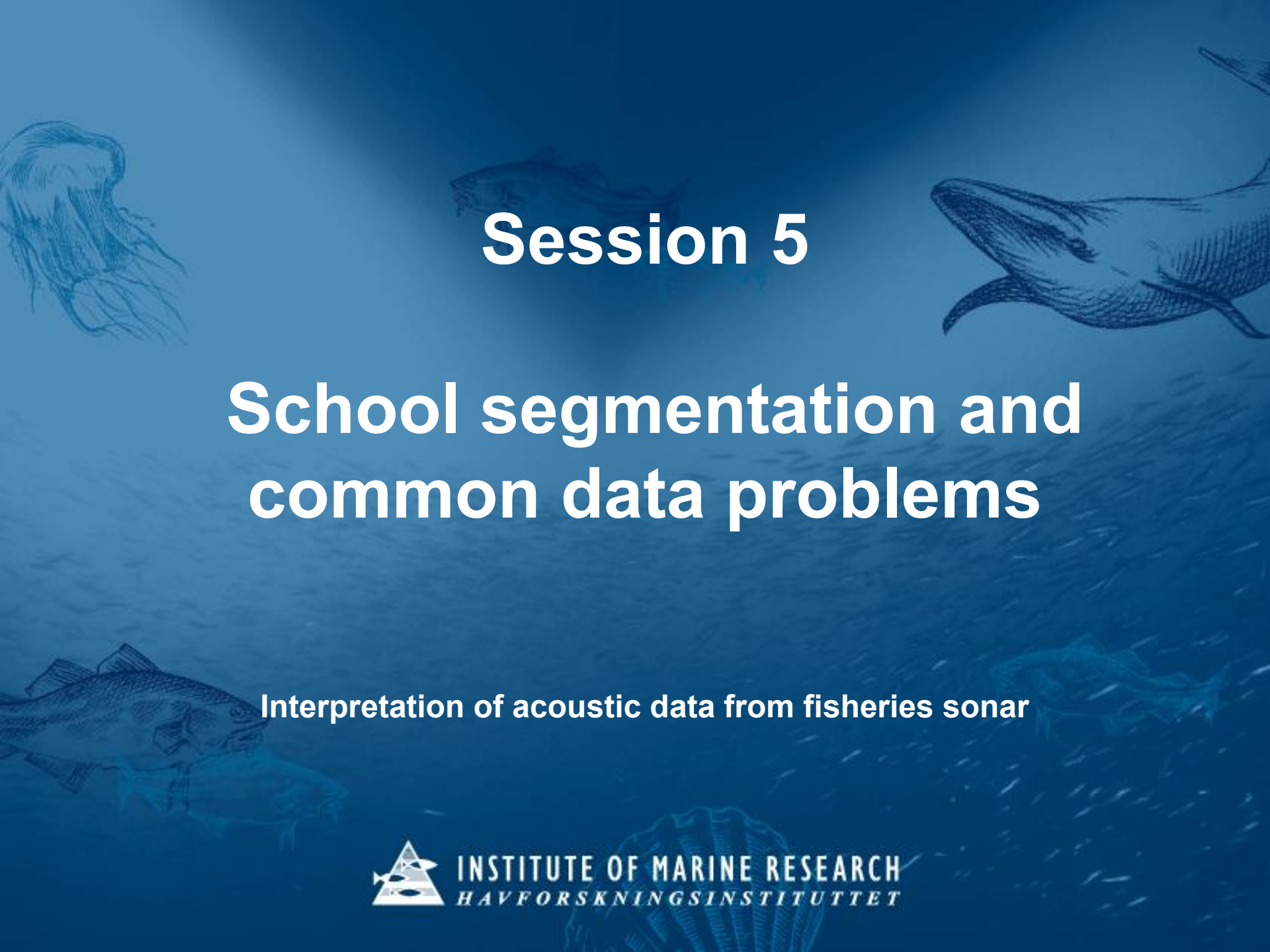




Session 5

School segmentation and common data problems

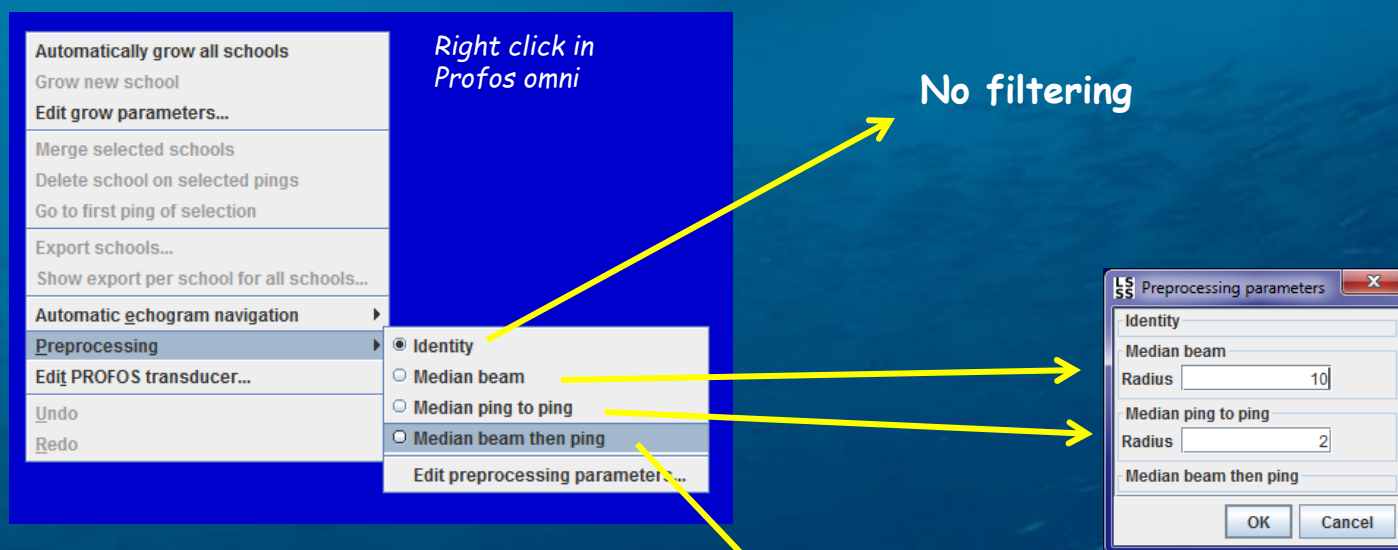
Interpretation of acoustic data from fisheries sonar



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Procedure for processing sonar data

1. Load sonar data
 - Select small data set (minutes) where there is indication of schools in sonar (sonar logg book, trawl data, echo sounder data)
2. Activate the smooth function (**Preprocessing** ⇒ **Median beam then ping**)

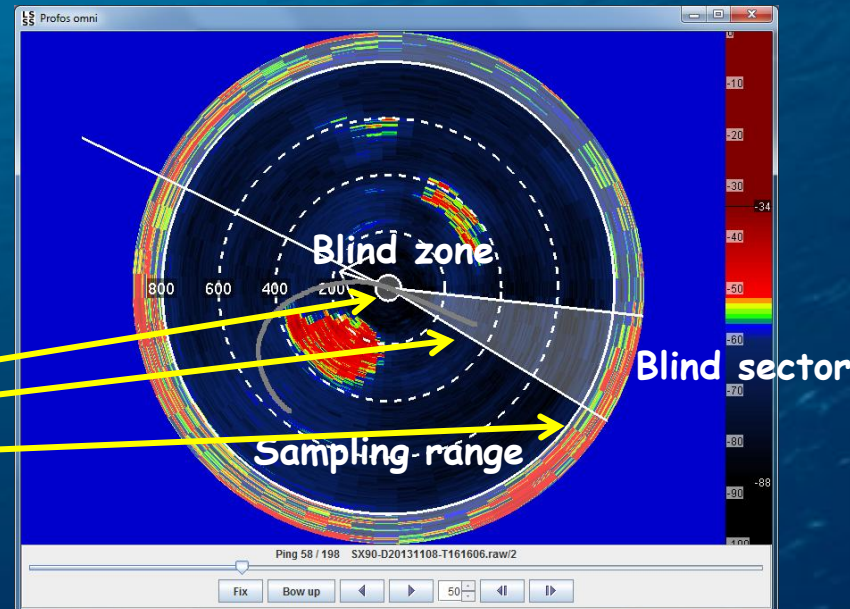


Median 21 samples along the beams
and median between 5 pings



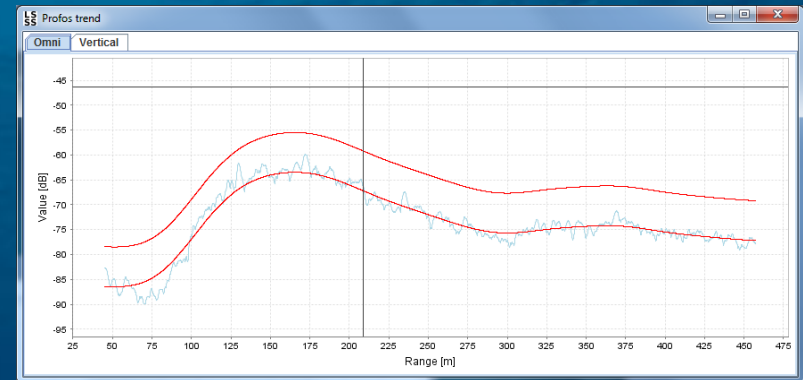
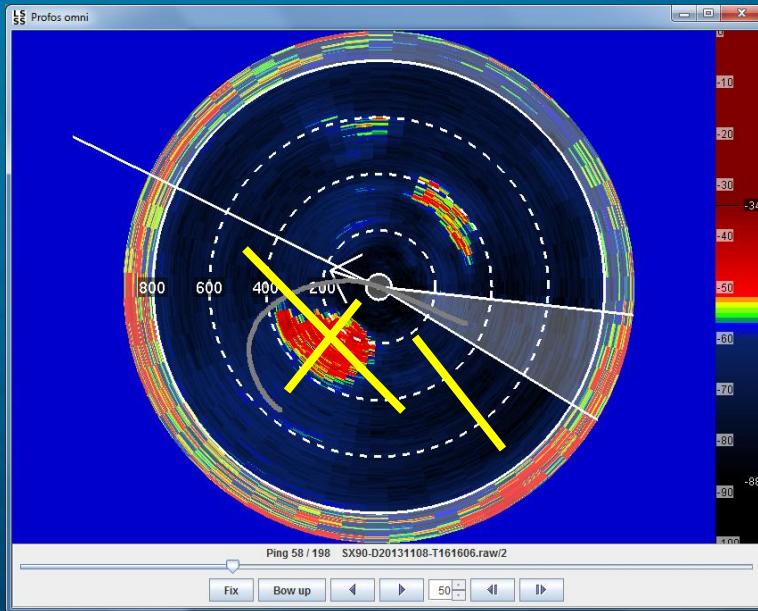
The diagram is a pyramid divided into four horizontal levels. From bottom to top, the levels are labeled: 'Producers', 'Primary Consumers', 'Secondary Consumers', and 'Tertiary Consumers'. Arrows indicate the flow of energy from the producers up to the tertiary consumers. The pyramid is shaded with a gradient from light at the bottom to dark at the top.

-
- The screenshot shows the 'LSSS Configuration file settings' dialog box. The 'PROFOS transducer' section is active, displaying the following settings:
- | Parameter | Value | Unit |
|-------------------------|-------|---------|
| Along ship coordinate | 0 | m |
| Athwart ship coordinate | 0 | m |
| Depth coordinate | 0 | m |
| Blind zone | 45 | m |
| Blind sector | 25 | degrees |
| Sampling range | 1200 | m |
| Vertical blind sector | 30 | degrees |
- The 'OK' button is highlighted with a yellow arrow.



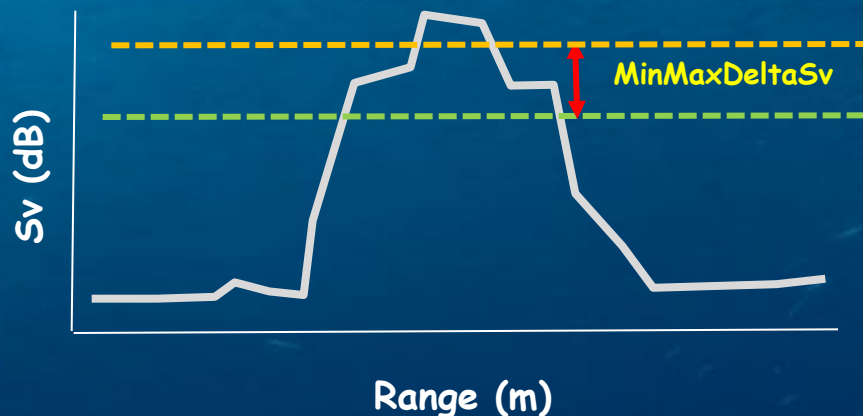
Procedure for processing sonar data

5. Inspect one ping with schools present, using **Profos trend windows**
6. Navigate image and define adaptative threshold (**Edit grow parameters** $\Rightarrow$ **Adaptative threshold**) suitable to separate background and school levels



Procedure for processing sonar data

7. Inspect how many pings schools are observed in the sonar (i.e. 3, small schools; 10, large schools). Change accordingly in grow parameters (**Edit grow parameters** $\Rightarrow$ **MinMaxFromSeed**)
8. If data has low number of missing or wrong pings, keep default value (**Edit grow parameters** $\Rightarrow$ **MinMissingPings** = 1)
9. Use default value of MinMaxDelta (**Edit grow parameters** $\Rightarrow$ **MinMaxDelta** = 2)



Procedure for processing sonar data

10. Use default value for minimum school area (**MinArea = 100 m²**, ca. 10 m diameter), maximum school area (**MaxArea = 10000 m²**, ca. 100 m diameter, maximum aspect ratio (**MaxAspecRatio = 10**, school length between beams maximum 10 times larger than school length along beam)
11. If small schools are detected few number of detections, the minimum number of pings the schools spans have to be reduced from default value (default: **MinPingCount = 7**)



Procedure for processing sonar data

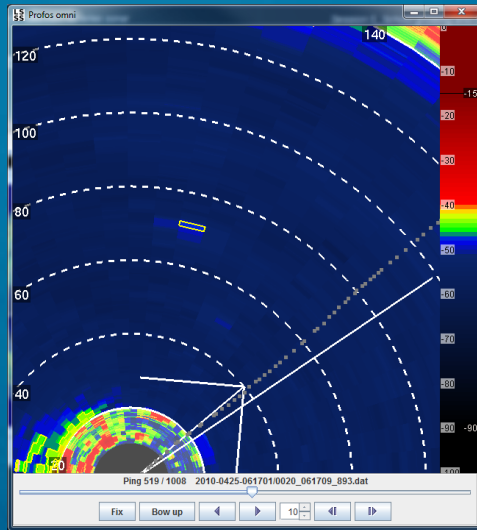
12. Manually segment some schools (Right click on top of school ⇒ Grow school)
13. Inspect results of manual procedure and adjust parameters
14. Start automatic school segmentation (Right click ⇒ Automatic grow all school)
15. Inspect results of manual procedure and adjust parameters

General criteria is to aim for schools with larger number of detections, better few "correct schools" than many "wrong schools"

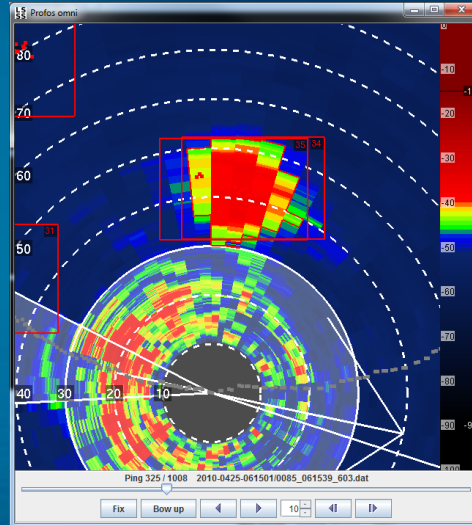


Code for rejected regions

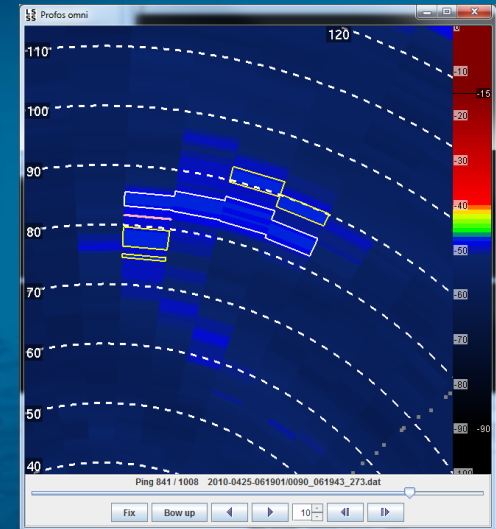
Yellow = area



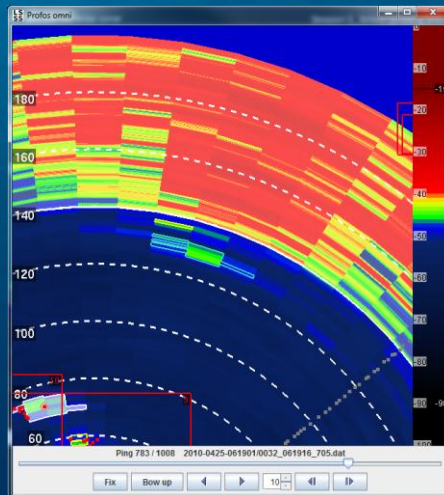
Red = blind region



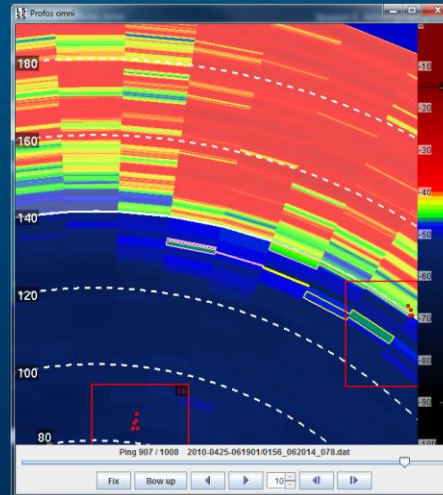
White = MinMaxSv



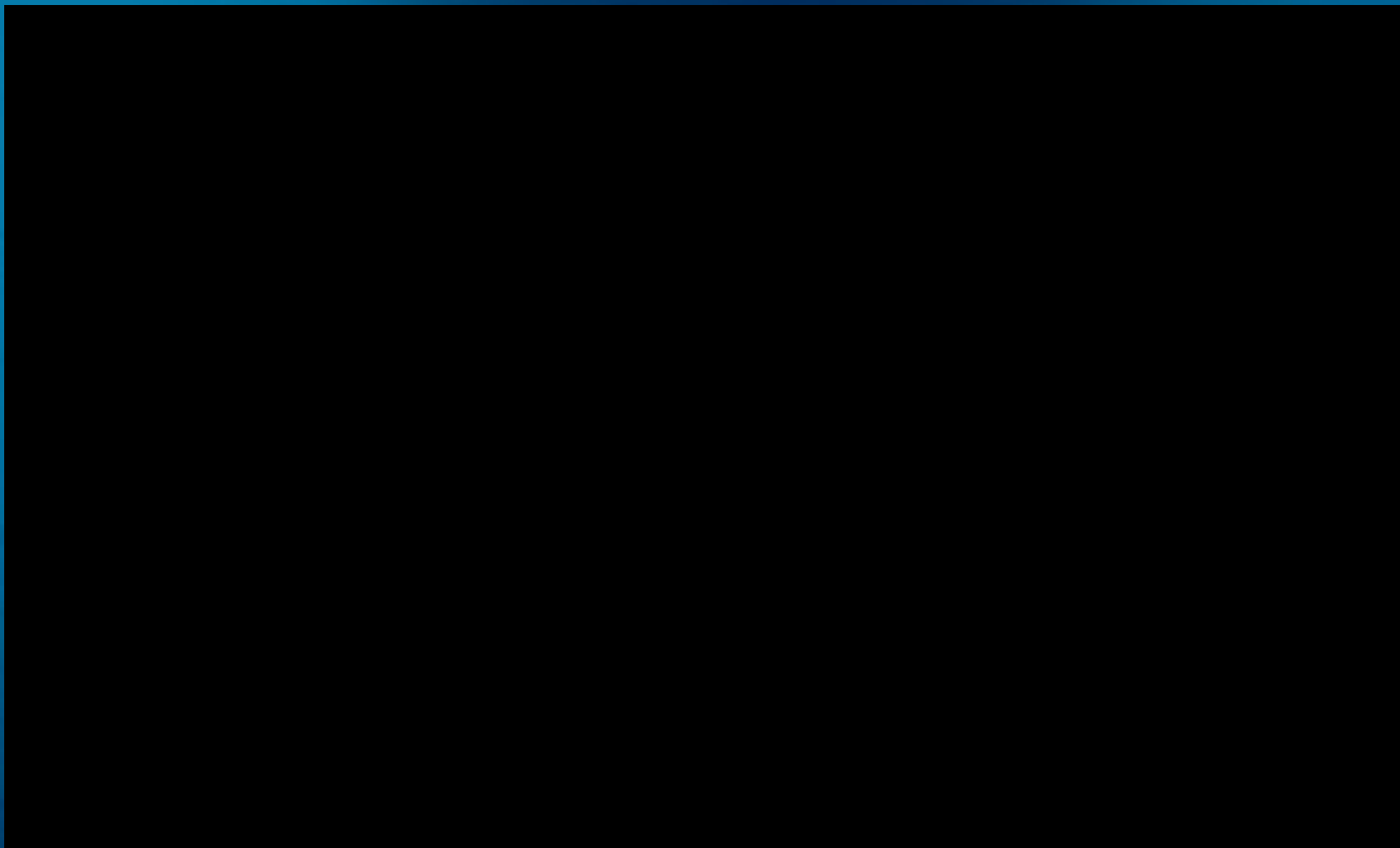
Green = pings span



Pink = Aspect ratio



Example of noisy data from surface reflection



Session_5_6

